ECEN 133 HW 9 Dylan Trembly

1.

a. $\begin{bmatrix}1&1&1\\1&1&1\\1&1&1\end{bmatrix} = \begin{bmatrix}1\\1\\1\end{bmatrix}[1\,1\,1]$ separable

b. $\begin{bmatrix}1&2&1\\2&4&2\\1&2&1\end{bmatrix} = \begin{bmatrix}1\\2\\1\end{bmatrix}[1\,2\,1]$ separable

c. $\begin{bmatrix}0&-1&0\\-1&4&-1\\0&-1&0\end{bmatrix}$ = not separable

d. $\begin{bmatrix}1&2&1\\0&0&0\\-1&-2&-1\end{bmatrix} = \begin{bmatrix}1\\0\\-1\end{bmatrix}[1\,2\,1]$ separable

e. $\begin{bmatrix}-1&-2&-1\\0&0&0\\1&2&1\end{bmatrix} = \begin{bmatrix}-1\\0\\1\end{bmatrix}[1\,2\,1]$

separable

f. $\begin{bmatrix}1&0&-1\\2&0&-2\\1&0&-1\end{bmatrix} = \begin{bmatrix}1\\2\\1\end{bmatrix}[1\,0\,-1]$ separable

2. $f(m,n) = \begin{cases}40 & n<20\\ 140 & n\geq 20\end{cases}$

18: $40\cdot\begin{bmatrix}1&2&1\\2&4&2\\1&2&1\end{bmatrix}$

19: $\begin{bmatrix}40&80&140\\80&160&280\\40&80&140\end{bmatrix}$

20: $\begin{bmatrix}40&280&140\\80&560&280\\40&280&140\end{bmatrix}$

21: $\begin{bmatrix}140&280&140\\280&560&280\\140&280&140\end{bmatrix}$

a.

$40(1+2+1+2+4+2+1+2+1) = 640$

$140(1+2+1+2+4+2+1+2+1) = 2240$

$g(m,n) = \begin{cases}640 & n\leq 18\\ 2240 & n\geq 21\\ 1040 & n=19\\ 1840 & n=20\end{cases}$

← offset by ±1 for n depending on where n is supposed to be. I choose the middle of matrix

b.

$g(m,n) = \begin{cases}0 & n\leq 18\\ 0 & n\geq 21\\ 100 & n=19\\ 100 & n=20\end{cases}$

$\begin{bmatrix}0&-1&0\\-1&4&-1\\0&-1&0\end{bmatrix}$ ← same process

c. $g(m,n) = \begin{cases}0 & n\leq 18\\ 0 & n\geq 21\\ -400 & n=19=20\end{cases}$

$\begin{bmatrix}1&0&-1\\2&0&-2\\1&0&-1\end{bmatrix}$

d. a smooths

b spikes a lot positively

c spikes a lot negatively

3. $T' = \frac{3}{8}T$ $\quad L=8$, $3=M$

$x(n) \rightarrow \boxed{\uparrow L} \rightarrow \boxed{H_u(z)} \xrightarrow{w(n)} \boxed{H_d(z)} \rightarrow \boxed{\downarrow M} \rightarrow x'(n')$

rate	$f_s = T$	$8f_s$	$w_c = \frac{\pi}{8}$	$8f_s$	$w_c = \frac{\pi}{3}$	$8f_s$	$\frac{8f_s}{3}$
int	T	$T/8$		$T/8$		$T/8$	$\frac{3T}{8}$

$\frac{\pi}{8}$ = norm freq (highest freq passed through filter)

$\frac{3\pi}{8}$ = norm freq of resampled (highest possible ω_0')

$\boxed{\omega_0' = \frac{3\pi}{8}}$

Input freqs suppressed: $\frac{\pi}{8}$ to π

4. $T' = \frac{5}{2}T$

$x(n) \rightarrow \boxed{\uparrow L} \rightarrow \boxed{H_u(z)} \xrightarrow{w(n)} \boxed{H_d(z)} \rightarrow \boxed{\downarrow M} \rightarrow x'(n')$

rate	$f_s = T$	$2f_s$	$w_c = \pi/2$	$2f_s$	$w_c = \pi/5$	$2f_s$	$\frac{2}{5}f_s$
in	T	$T/2$		$T/2$		$T/2$	$5/2T$

$\frac{\pi}{2}$ = norm freq

$\frac{5}{2}\pi$ = norm freq of resampled

input freqs suppressed $\frac{\pi}{5}$ to π

$\boxed{\omega_0' = \frac{\pi}{5}(5) = \pi}$

5. $h = [1\ 4\ 6\ 4\ 1]$

a.

$y(n) = x(n)$ convolve $h(n)$

$y(n) = x(n-2) + 4x(n-1) + 6x(n) + 4x(n+1) + x(n+2)$

$x_s(n) = x(n-1)$

$x_d(m) = x(2m)$

$x_{sd}(m) = x_s(2m)$

b. $x_d(m) = x(2m)$ $\quad h = [1\ 6\ 1]$

$y_1(m) = x_d(m-1) + 6x_d(m) + x_d(m+1)$

$y_1(m) = x(2m-2) + 6x(2m) + x(2m+2)$

$y_2(m) = 4x_{sd}(2m) + 4x_{sd}(2(m+1))$

$= 4x(2m-1) + 4x(2m+1)$

c. $y_1(m) + y_2(m) =$

$x(2m-2) + 4x(2m-1) + 6x(2m) + 4x(2m+1) + x(2m+2)$

Same as a where $2m = n$

d. These filters cover the full original filter $[4\ 4]$, $[1\ 6\ 1]$ cover $[1\ 4\ 6\ 4\ 1]$. All indexes are filtered only once